

KWANELE NTSHANGASE

0789325315 | [Portfolio](#) | [Email](#) | [GitHub](#)

Education

Nelson Mandela University

Diploma in Information Technology

— Dec 2025

Skills

Programming Languages: C#, JavaScript, TypeScript, Python, LINQ, CSHTML

Databases: SQL Server, PostgreSQL, Supabase, MySQL, Azure Database

Frameworks & Libraries: ASP.NET Core, React, Node.js, Express, .NET

Developer Tools: Git, Docker, Power BI, Excel, Zapier, Power Automate

Cloud & DevOps: Azure, CI/CD, Render, Vercel

Soft Skills: Critical Thinking, Problem-Solving, Effective Communication, Leadership

Languages: English, IsiZulu, IsiXhosa

Projects

[ScrapFlow SA](#) | *Full-Stack Scrapyrd System*

— March 2026

- Built and deployed a full-stack scrapyrd management platform using ASP.NET Core 8 (Clean Architecture, EF Core), React 18, and PostgreSQL, implementing a 6-step digitised inbound ticket workflow (vehicle arrival → weighing → grading → tare → payment → completion) that eliminates paper-based processing across multiple yard sites.
- Engineered real-time inventory tracking using ASP.NET Core SignalR WebSockets, broadcasting live stock updates to all connected React clients on ticket completion — removing the need for manual page refreshes and enabling accurate multi-site inventory visibility without polling
- Implemented a 5-role RBAC system (Owner, Manager, ScaleOp, Cashier, Viewer) secured with JWT bearer authentication and ASP.NET Core authorization policies, deployed to a cloud production environment (Render + Vercel) with GitHub Actions CI/CD pipelines automating build validation on every pull request.
- Designed a webhook automation engine using HMAC-SHA256 signed HTTP payloads that integrates with third-party platforms (Zapier, n8n) to trigger supplier payment receipts and manager alerts on business events — with a React/Zustand Owner-only management UI for creating, testing, and toggling webhook subscriptions without code changes.

[WardCare+](#) | *Final Year Project*

— Oct 2025

- Solely owned the cloud deployment pipeline for a 4-person team project: provisioned Azure App Service and Azure SQL Database, configured GitHub Actions CI/CD with OIDC workload identity federation (zero stored credentials), and automated EF Core database migrations on startup — reducing every release to a single push to main.
- Collaborated in a team of 4 to architect and deliver a production-grade ASP.NET Core 8 MVC application with 7 RBAC roles, policy-based authorization, FluentValidation, and a service-layer pattern (IPatientService, IVitalSignService) — applying Git branch workflows throughout to enable parallel development across subsystems.
- Designed and delivered the Doctor–Patient subsystem end-to-end — C# MVC controllers, Razor views, EF Core 8 queries, and Bootstrap 5 UI — including in-memory PDF generation (QuestPDF) for downloadable clinical patient folders, contributing to a live multi-role application still accessible at the deployed URL.
- Implemented a POPIA-compliant audit trail using an EF Core SaveChangesInterceptor to capture before/after JSON snapshots on every INSERT, UPDATE, and DELETE across 25+ entities, combined with a soft-delete pattern on all tables — demonstrating applied knowledge of real-world data compliance requirements, not just framework syntax.

- Architected and deployed a production-grade REST API using ASP.NET Core 8 and Clean Architecture (3-layer: API / Core / Infrastructure), exposing 35+ endpoints across 7 controllers with JWT authentication, refresh token rotation, role-based access control (Admin, Librarian, Patron), and fixed-window rate limiting — hosted on Azure App Service with PostgreSQL 17 (Azure Flexible Server) and automated Entity Framework Core migrations on startup
- Built a real-time notification system using SignalR WebSockets with group-based routing and a daily background service that probes active loans for due-date alerts; resolved a token refresh race condition on the frontend using a subscriber queue pattern, eliminating duplicate refresh calls across parallel Axios requests
- Developed a responsive React 19 SPA (Vite, Tailwind CSS) with Recharts analytics dashboards, QR code loan management (ZXing.Net), voice search via the Web Speech API, and a multi-step password reset flow backed by database-persisted, cryptographically secure 6-digit tokens (RandomNumberGenerator) — deployed to Vercel with automatic CI/CD on every GitHub push
- Eliminated N+1 database queries by refactoring repository interfaces to use eager-loaded GetAllAsync() with .Include(), implemented a GlobalExceptionHandler returning structured ProblemDetails JSON with correlation IDs, and added PostgreSQL health check endpoints (/health/ready, /health/live) used as Azure platform liveness and readiness probes.

Certifications

C# Programming Award

Nelson Mandela University

Student Merit Awards(2024 &2025)

Nelson Mandela University

Introduction to Cyber Security Certificate

Cisco

Data Analyst BootCamp: Python, Excel, Power BI & SQL

Udemy

Angular 20 & ASP.NET Core Web API

Udemy